

Information Technology (IT) – Student Notes



1. Introduction to Computers

- **Definition:** An electronic device that processes data to produce information.
- **Main Components:**
 - **Input Devices:** Keyboard, Mouse, Scanner
 - **Output Devices:** Monitor, Printer, Speakers
 - **Processing Unit:** CPU (Central Processing Unit)
 - **Storage:** Hard Disk, SSD, Pen Drive



2. Types of Computers

- **Supercomputer**
- **Mainframe**
- **Mini Computer**
- **Microcomputer (PC)**



3. Software

- **System Software:** Windows, Linux, macOS
- **Application Software:** MS Word, Excel, Tally
- **Utility Software:** Antivirus, Disk Cleanup
- **Programming Languages:** C, Java, Python



4. Internet Basics

- **Browser:** Chrome, Firefox, Edge
- **URL:** Uniform Resource Locator (e.g., <https://www.google.com>)
- **Search Engine:** Google, Bing, Yahoo
- **Email:** Electronic Mail (e.g., Gmail, Outlook)



5. MS Office Tools



MS Word

- Create, edit, format text documents
- Save as .docx, .pdf
- Shortcut: Ctrl + B (Bold), Ctrl + S (Save)

MS Excel

- Used for data handling, formulas
- Common formulas: =SUM(), =AVERAGE()
- Save as .xlsx

MS PowerPoint

- Create presentations with slides
- Use of animations and transitions
- Save as .pptx

6. Computer Networks

- **LAN (Local Area Network)**
- **WAN (Wide Area Network)**
- **Internet:** Largest WAN
- Devices: Router, Modem, Switch

7. Cyber Safety & Security

- Use strong passwords
- Don't share personal data
- Beware of phishing emails
- Use antivirus software

8. Number Systems

- **Binary (Base-2)**
- **Decimal (Base-10)**
- **Octal (Base-8)**
- **Hexadecimal (Base-16)**



9. Data and Memory

- **Bit:** Smallest unit of data
- **Byte:** 8 bits
- **KB, MB, GB, TB:** Units of storage
- **RAM:** Temporary memory
- **ROM:** Permanent memory



10. IT Applications in Daily Life

- **E-Governance**
- **Online Banking**
- **E-Commerce**
- **Telemedicine**
- **E-Learning**



Types of Programs That Run on a Laptop

1. System Software

- Controls the basic functions of the laptop.
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- **Examples:**
 - **Operating Systems:** Windows, macOS, Linux
 - **Device Drivers:** Graphics driver, sound driver
 - **Utilities:** Disk Cleanup, Antivirus
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2. Application Software

- Helps users perform specific tasks.
- **Examples:**

- **MS Office:** Word, Excel, PowerPoint
 - **Web Browsers:** Google Chrome, Firefox
 - **Media Players:** VLC, Windows Media Player
 - **PDF Readers:** Adobe Acrobat Reader
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3. Programming Software

- Used for writing and testing code.
 - **Examples:**
 - **IDEs:** Visual Studio, PyCharm, Eclipse
 - **Text Editors:** Notepad++, Sublime Text
 - **Compilers:** GCC, Java Compiler
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4. Graphics and Design Software

- Used for photo editing, video editing, or design.
 - **Examples:**
 - **Adobe Photoshop, Illustrator**
 - **CorelDRAW**
 - **Canva (web-based)**
 - **Filmora, Adobe Premiere Pro** (video editing)
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5. Communication Software

- Used to communicate over the internet.
- **Examples:**
 - **Zoom, Microsoft Teams**
 - **Skype, Google Meet**
 - **Email clients:** Outlook, Thunderbird

6. Gaming Software

- Games and game launchers.
 - **Examples:**
 - **Steam, Epic Games Launcher**
 - Popular games like **GTA V, Minecraft, Valorant**
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7. Database Software

- Used to manage and operate databases.
 - **Examples:**
 - **MySQL**
 - **Oracle Database**
 - **Microsoft Access**
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8. Security Software

- Protects your laptop from malware.
- **Examples:**
 - **Antivirus:** Quick Heal, Norton, Avast
 - **Firewall**
 - **VPN software**